

Project 0: Bicycle World (Linux Introduction)

- Due Jan 14 by 11:59pm
- Points 75
- Submitting an external tool

Welcome to Bicycle World, the premier text-based bicycle shop! This shop is only accessible to programmers like you who are familiar with the command line. In this project, you will use Unix commands to navigate and edit the filesystem of this special shop.

Goals:

- Use Unix commands to learn about the company Bicycle World.
- Gain beginner knowledge of Unix commands.
- Learn how to control your computer's basic movements without using a mouse.

Takeaway:

By the end of this project, you will be able to navigate, access, and modify files and directories on your computer. Feel free to use the man pages or Google for commands, as some of them will be used in future assignments. :)

Prerequisite:

1. In your Linux VM, install Git:

```
sudo apt install git
```

2. Ensure you're in your home directory:

```
cd ~
```

3. Log in to GitHub using the GitHub CLI

1. Install GitHub CLI

```
sudo apt update  
sudo apt install gh
```

2. Authenticate using the CLI by following the on-screen prompts (**USE GITHUB.COM AND HTTPS**)

```
gh auth login
```

4. Clone your newly created Git repository from GitHub:

- Go to your GitHub repository page and click "Clone."
- Copy the link under "Clone with HTTPS" and run:

```
git clone <https://github.com/<your-username>/cy2550.git>
```

5. Enter your class directory:

```
cd cy2550/
```

Instructions

Screenshot 1:

- Make and enter the project0 directory
- Create a file called 'hello.txt' and this file **MUST** contain your full name and Github username exactly in this format:

```
Hi, my name is Mohit Singhal, and my Github user name is Mohit Singhal.
```

- Download the bicycle-world zip folder

```
wget https://s3.amazonaws.com/bucket.sakibnm.space/cy2550/bicycle-world.zip
```

- Extract the zip file:

```
unzip bicycle-world.zip
```

- Do an 'ls' to check if you have the extracted directory "bicycle-world" in the project directory.

Screenshot 2:

- Print the date in the command line. (HINT: 'date' is a command.)
- Print the current working directory.

Screenshot 3:

- Change directories to the freight/ directory.
- List the files and directories in this directory.

Screenshot 4:

- Get into the porteur/ directory.
- Change directories by going down two levels to the bicycle-world/ directory using a single command line.

Screenshot 5:

- Change directories to the mountain/downhill directory using a single command.

Screenshot 6:

- Make a file called dirt.txt.
- Make a file called mud.txt.
- List the files and directories in the downhill/ directory with their corresponding permissions.

Screenshot 7:

- In the downhill/ directory, make a directory called safety/.
- Create a file named 'safety-notes.txt' in the directory.
- Put the following sentence, "Do not paddle and use brakes while riding downhill." to the 'safety-notes.txt' file.
- Use the commands 'echo' and '>' to do that. Do not use any editors like 'nano' or 'vim'.

Screenshot 8:

- Create a file titled 'safety-notes-part2.txt' and modify the permissions of this file such that all users can execute it.
- When the file 'safety-notes-part2.txt' is executed (./safety-notes-part2.txt), it should display the following output on your terminal:

```
Oh hey there, I am glad you made it here, please be sure to wear your helmet while biking on the road.
```

HINT: you must use 'echo' command and use a shebang (`#!/usr/bin/bash`) to do that

- Create a new folder directory named "Noted". Then, make a new copy of "safety-notes-part2.txt" named "read-notes.txt" and move the "read-notes.txt" into the "Noted" folder.

Submitting Your Project

Step 1: Configure Git

Set up Git with your name and email:

```
git config --global user.name "your-username"  
git config --global user.email "your-email@example.com"
```

Step 2: Commit and Push Changes

1. Navigate to `project0/`:

```
cd ~/cy2550/project0
```

2. Add all files:

```
git add .
```

3. Commit your changes:

```
git commit -m "Initial commit for project 0"
```

4. Push to GitHub:

```
git push
```

5. Verify your files are in your GitHub repository.

Step 3: Submit to Gradescope

Submit your GitHub repository link via Gradescope, accessible through the course's Canvas page.

This tool needs to be loaded in a new browser window

Load Project 0: Bicycle World (Linux Introduction) in a new window